BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 133 · LANs, WANs and Network Models

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Convert 00101101 binary to denary.

Q2. What is the Internet?
A. A network of networks
B. One central web server
C. Only public web pages
D. One school's wireless LAN

Q3. “A web client must host every website before it can request a page.” Correct this. Explain why.

Q4. A pupil needs work on several devices. “Cloud files are always available when the Internet fails.” Correct this. What could happen?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LAN — Covers a small geographical area such as one building, usually under a single organisation.

WAN — Connects networks over wider geographical areas, often via third-party infrastructure.

Client-server — Central servers provide resources/services for client devices.

Peer-to-peer — Devices share resources directly without a dedicated central server for those resources.

New learning

LAN: Covers a small geographical area such as one building, usually under a single organisation.

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Worked example

A company has two offices in different cities connected so staff share files. Identify network types.

1. Each office can use its own LAN.

2. A WAN connects geographically separate LANs.

3. A server can centrally control shared files and access.

4. Explain why peer-to-peer may be less suitable for many managed users.

We Do · Practise with guidance

Guided 1

Compare a five-computer home network with a national retailer's network.

Worked method: Home is LAN; retailer uses LANs connected by WAN.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the missing output for a connection between two distant school sites.

```
separateSites = true
if separateSites then
    ...
else
    print("One LAN")
endif
```

write

Question 1

Explain one difference between LAN and WAN. [2 marks]

Question 2

Give one advantage of client-server networking. [2 marks]

Question 3

Explain one drawback of peer-to-peer for a large school. [3 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A WAN can connect LANs in geographically separate locations.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 45, because 32 + 8 + 4 + 1 = 45.

Q2. A — A network of networks

Q3. A server stores and makes website resources available for clients to request.

Q4. Cloud resources are accessed over a network. Files may be unavailable if the connection or server fails.

I Do / We Do pages · Guided practice

Q1. Home is LAN; retailer uses LANs connected by WAN.

You Do pages · Code First

```
separateSites = true
if separateSites then
    print("WAN connects sites")
else
    print("One LAN")
endif
```

Expected result: WAN connects sites

You Do pages · Independent questions

Q1. LAN covers smaller/local area; WAN connects over wide geographical areas.

Q2. Centralised user account, access or resource management.

Q3. Harder to centrally manage permissions, security or backups across many devices.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: